## (i) Source run, $M(p_{\text{src}}) \Rightarrow c$

```
   ...
a  (1) whether the number is prime
c  (2) the number of set bits
b  (3) whether the number is positive
   ...
   <code>
   def func(n: int):
     a x = n > 1 and all(...)
     b y = (n > 0)
→    c z = bin(n).count('1')
```

$q_{\text{src}}$ goal-head queries at the source pivot (carry the source progress)

## (ii) Clean destination run, $M(p_{\text{dst}}) \Rightarrow b$

no intervention: the goal heads read b, the next listed task

```
   ...
a  (1) whether the number is prime
b  (2) whether the number is positive
d  (3) the last decimal digit
   ...
   <code>
   def func(n: int):
     a x = n > 1 and all(n % i != 0 ...)
       y = n > 0
```

gaze: 0.07 · **0.24** · 0.06

attention × |v| from the pivot, mean over goal heads

*"I'm checking if the number is greater than zero."*

GoalScope readout at the pivot

## (iii) Select, $M(p_{\text{dst}} \mid \leftarrow q_{\text{src}})$

transfer the source query; read which task the goal heads attend to

```
   ...
a  (1) whether the number is prime
b  (2) whether the number is positive
d  (3) the last decimal digit
   ...
   <code>
   def func(n: int):
     a x = n > 1 and all(n % i != 0 ...)
       y =
```

gaze: 0.05 · 0.07 · **0.31** → d

*"I'm checking the last digit of the number by using the modulus operator (` % 10 `)."*

GoalScope readout at the pivot (readout-shift)

## (iv) Restrict and Generate, $M(p_{\text{dst}} \mid \text{mask}_{\text{d}}) \Rightarrow d$

fresh run, no query transfer; goal heads may read only d

```
   ...
a  (1) whether the number is prime
b  (2) whether the number is positive
d  (3) the last decimal digit
   ...
   <code>
   def func(n: int):
     a x = n > 1 and all(n % i != 0 ...)
→    d y = n % 10
```

**From the pivot on, goal heads may attend only to**

- the selected task's span (d)
- the pivot token " ="
- the tokens generated so far
- all other keys are masked (pre-softmax)

Non-goal heads read the full context.